Mathematical overview: fewest and indispensable are different

The model. An $n \times n$ square grid has an open boundary. Only side-neighbors count; there are no outside cells or wraparound edges. Choose seeds, then repeatedly fill any empty cell touching at least two filled side-neighbors. Filled cells stay filled. Stop only when no legal addition remains. A partner checks each addition.

Every completed order has the same finish. The board is finite and the filled set grows. Let C be any closed set containing the seeds, where closed means no cell outside C has two side-neighbors in C. Inductively every legal addition belongs to C. By that induction a terminal run lies inside every closed superset of the seeds, and it is itself closed, so it is the least one. Every terminal run is thus the same least closed superset. Different intermediate pictures or timings do not contradict this final-state result.

Exactly n seeds are needed to fill an $n \times n$ board. Count exposed unit sides, including the board boundary and any hole boundaries. Adding a cell with k already filled side-neighbors changes this count by $4 - 2k$. Legal additions have $k \geq 2$, so perimeter never increases. A start of s seeds has at most $4s$ exposed sides. The full board has $4n$, forcing $s \geq n$. Conversely the n diagonal cells fill: induct on the distance $|r - c|$ from the main diagonal. Each cell in the next diagonal layer has two neighbors in the previous layer. This supplies both an impossibility certificate and an attaining construction.

Every seed can be necessary in a non-smallest start. Inclusion-minimal means removing any seed prevents filling; minimum means no filling start has fewer seeds. The four seeds $(1,1), (1,2), (2,3), (3,3)$ (row, column, counted from 1 at the top left) on a 3×3 board are all indispensable, yet three diagonal seeds fill. Neither notion implies that all legal growth orders have the same intermediate picture.

What children discover. The Grades 2–3 route emphasizes choosing, growing, resetting and comparing small seed sets. Grades 4–5 adds perimeter as a lower bound and the distinction between local removal tests and a global optimum. Reading can be supported; the child should retain the choice of seeds and legal additions. Counting exposed edges and interpreting a general n are later readiness demands. No K–1 packet is claimed; the preparation section gives an adult-led K–1 route on the Grades 2–3 pages.

Evidence and limits. A completed construction proves one start works. Exhaustive checks prove the stated small-board counts. The closure, perimeter and diagonal arguments prove the general assertions. This is deterministic bootstrap percolation, not a lesson proving a random critical threshold. Familiar closure and perimeter ideas are deliberately reused for a new optimization question.

Preparation, shared launch and pacing

Status. Prepared for organizer review; unpiloted. Physical printing, preparation, handling and timings remain unrehearsed.

K–1 route, adult-led. No K–1 packet is printed; a K–1 table uses Grades 2–3 pages, with the parent reading and recording. Read the rule as one sentence: "A square gets a counter when two squares next to it, not corner to corner, have counters." After the launch, use Problem 1 (page 1): the child places three seeds, grows them while a partner points to the two neighbors, and the parent asks, "Did it fill, or did it get stuck?" Then put counters on Problem 2's large 4×4 board: "Can you fill it with fewer seeds?" The parent draws each start that fills on a small board. End with Problem 5 (page 5): each child grows one of the four small boards, and the table compares. Key in plain words: three seeds in a slanting line fill the 3×3 board, and three in a straight row get stuck at once; four seeds can fill the 4×4 board, and three never can; every small board on page 5 gets stuck (6, 3, 3 and 6 squares), so the big start needs all four seeds, even though three slanting seeds fill the board. That K–1 children keep the two-neighbor rule for a session is a prediction.

Per pair. Sixteen counters or square tiles no wider than 25 mm, in two colors: one color for seeds, the other for added squares. The packets' 30 mm boards are the workboards: the 3×3 boards on Grades 2–3 pages 1, 3, 4 and 5 and Grades 4–5 pages 1, 2 and 7, and the 4×4 board on Grades 2–3 page 2 and Grades 4–5 page 3. No separate board or mask is needed. Keep a reset sketch visible. For 5×5 work supply 25 counters or use pencil diagrams. Short sticks or drawn strokes can mark exposed edges later. Pieces must not conceal neighboring cells; exact tile-to-cell fit is unnecessary. Print US Letter, single-sided at Actual Size. Smaller printed grids are records. For the launch, draw a 2×3 grid of 30 mm squares on paper or a whiteboard.

Staffing. Two pairs at the middle table and a pair plus rotating checker at the upper table. An adult reads as needed, but the partner should identify the two filled side-neighbors before a new counter is added. Rotate seed designer, grower and checker. Counting counters into bags for three pairs and a demonstration set takes about 10 minutes beyond printing; this estimate needs a real rehearsal.

Launch after the movement opening. Let children handle the counters for two minutes. On a separate 2×3 board use the input below. Fill the top middle, pointing to its left and lower neighbors. Neither right-hand cell can then fill. Diagonal touching alone is insufficient. The start stays marked while later counters are added.

Input	One legal addition	Finish
$S \;\; \cdot \;\; \cdot$	$S \;\; + \;\; \cdot$	$S \;\; + \;\; \cdot$
$S \;\; S \;\; \cdot$	$S \;\; S \;\; \cdot$	$S \;\; S \;\; \cdot$

S is an original seed and $+$ a later addition. This example teaches a whole run and stopping without supplying an optimal square start.

A first visit. After the five-minute movement opening and five-minute launch, allow 20–25 minutes for contrasting starts and the first fewest-seed search. Use the remaining 10–15 minutes to compare different successful starts or test the same start with different legal choices. Close with one construction and one question. The perimeter proof and indispensable-seed investigation can fill another visit; do not rush every page into one hour. First route: Grades 2–3 Problems 1, 2 and 4, Grades 4–5 Problems 1–3. Return visit: Grades 2–3 Problems 3 and 5, Grades 4–5 Problems 4–6.

Hints only when useful. If a child changes the rules, ask the checker to touch the two supporting squares. If the original seeds are lost, reset from the sketch before comparing counts. When a search suggests an optimum but cannot establish it, ask what a filled board must have that a start cannot gain. With Grades 2–3, introduce perimeter sticks then, rather than prescribing a proof before any exploration; Grades 4–5 meet exposed sides in Problem 4. Adult recording is ordinary support; adult seed selection or legal-move selection is a signal to reduce the board.

Grades 2–3: constructions, finishes and reset tests

Coordinates below are $(\text{row}, \text{column})$, counted from 1 at the top left. They are adult lookup notation, not an extra student recording requirement.

Problem 1, page 1: three seeds that fill or stop

A main diagonal $(1,1),(2,2),(3,3)$ fills all nine cells. A full top row $(1,1),(1,2),(1,3)$ stops immediately: every empty cell has at most one filled side-neighbor. Many other choices work. Require the original three seeds to stay marked, and distinguish "I stopped" from "no legal addition remains." A counterexample to filling is a finished legal run, not simply a picture with empty squares.

Problem 2, page 2: fewest on 4×4

Four seeds suffice and three never suffice. The lower-bound proof is in the mathematical overview and developed below; the middle-table first success can be a convincing construction without immediately completing the proof. Three different four-seed starts are the main diagonal, the antidiagonal, and

$$\{(1,1),(1,3),(2,4),(4,1)\}.$$

The last start grows to the full board under the stated rule; it is independently checked in the package. Rotated or reflected placements are different starts unless an adult deliberately introduces a new symmetry-equivalence question. Do not silently reject them. The search is not asking for all optimal starts.

Problem 3, page 3: specified finish sizes

The following three-seed starts realize the four requested sizes:

Filled cells	Seeds	Terminal shape
3	$(1,1),(1,2),(1,3)$	the top row
4	$(1,1),(1,2),(2,1)$	upper-left 2×2 square
6	$(1,1),(1,2),(2,3)$	top two rows
9	$(1,1),(2,2),(3,3)$	the full square

No other total occurs: the 84 three-seed starts finish with 3, 4, 6 or 9 filled squares only (14, 16, 40 and 14 starts). The classification under Grades 4–5 Problem 1 explains why. Children can keep one seed sketch beside one physical board instead of building all four at once. The requested outputs contrast a stable line, a filled small square, a rectangle and the full board. They are not four copies of one calculation.

Problem 4, page 4: different legal orders

No completed runs from the same seeds finish differently. The least-closed-superset proof in the mathematical overview explains why. An explanation a child can follow: suppose run A added a square that run B never added, and take the first such square in A's order. When A added it, two of its neighbors were already filled, by seeds or by earlier additions of A, and B has all of those too. So that square still has two filled neighbors when B stops, and B was not finished. Every square one complete run adds, the other adds too. The numbers children write show the orders differ while the finishes agree. Diagonal seeds fill the board in every order; $(1,1)$, $(1,2)$, $(2,3)$ stop early, can begin with $(1,3)$ or $(2,2)$, and finish with the top two rows either way. Children write each number in a corner of its cell, in place of $+$, before the counter goes down. If a pair claims a counterexample, compare the seed sketches and look for an unfinished legal addition before correcting their conclusion.

Problem 5, page 5: every seed needed, yet too many

The seeds are $(1,1)$, $(1,2)$, $(2,3)$ and $(3,3)$. One legal growth sequence adds $(1,3)$ and $(2,2)$, then $(2,1)$ and $(3,2)$, then $(3,1)$. Each listed layer may be filled in either order because its supports are already present. These are adult verification layers, not a required simultaneous-round record.

Removing the four seeds in that order leaves terminal counts $6, 3, 3, 6$. Thus every seed is indispensable in this particular start. Four is nevertheless not the fewest: the three-seed diagonal fills. Let children share the four removal trials and pool their reset pictures. The labels *inclusion-minimal* and *minimum* can follow their discovery; no vocabulary quiz is needed.

Grades 4–5: constructions and an impossibility certificate

Problem 1, page 1: all finish sizes from three seeds

Before Problem 1, ask Grades 2–3 Problem 4's question aloud: partners place seeds $(1,1)$, $(1,2)$, $(2,3)$, one first adds $(1,3)$ and the other $(2,2)$, and they compare the finished boards (both are the top two rows; diagonal seeds, which fill in every order, would test nothing). Problem 1's "final numbers" assume each start has one finish; this settles it by experiment, and the overview's proof explains it.

Exactly four totals occur: $3, 4, 6, 9$. The preceding page supplies witnesses. The checker examines all $\binom{9}{3} = 84$ three-seed placements, including rotations and reflections, and finds no other totals. This finite check supports the adult answer; children need not list 84 cases individually.

A short geometric explanation can replace exhaustive listing. If there is no legal addition, the total is 3. Otherwise two seeds support a first addition. If those two seeds are diagonal neighbors, their closure is the 2×2 square they span. The third seed is inside it (finish 4), in an extra row or column next to it (finish 6), or diagonally beyond it (finish 9). If the supporting seeds are opposite across the new cell, their closure is a full row or column of the 3×3 board. A third seed in the next parallel line expands to a 2×3 rectangle, giving 6; one two lines away first creates a full perpendicular line and then fills the square, giving 9. These descriptions include reflected and rotated cases. If children are still searching, keep this classification with the adult.

Problems 2–3, pages 2–3: 4×4, 5×5 and 10×10

The minima are 4, 5 and 10 respectively. Main or opposite diagonals attain them. Problem 3 now asks for the fewest seeds on both larger boards, using drawn marks rather than counters. A successful construction establishes an upper bound; the perimeter argument supplies the matching lower bound. A larger successful start is useful experimental evidence but does not yet answer the minimum question.

For a diagonal proof, cells with $|r - c| = 1$ each touch two seeds. Once all cells at distance at most j from the diagonal are filled, a cell at distance $j + 1$ touches two cells at distance j. At an edge these two supports still lie on the board. Induction fills every diagonal layer. This establishes a construction for arbitrary n, not merely the printed sizes.

Problem 4, page 4: what an addition does to the boundary

Count *unit sides*, not cells, corners or the outside rectangle alone. An added cell shares k sides with filled cells: those k old exposed sides disappear and $4 - k$ new exposed sides appear. Thus the net change is $4 - 2k$. For $k = 2, 3, 4$ the changes are $0, -2, -4$. All legal additions have one of those values, so none increases perimeter. In the three printed pictures the before/after counts are $8 \to 8$, $12 \to 10$ and $16 \to 12$, respectively. Count them directly against the pictures when introducing the argument.

A hole boundary counts too: filling an isolated one-cell hole removes four exposed edges. Touching only at corners does not cancel exposed sides. If children count the perimeter of a surrounding rectangle, ask them to trace every filled-to-empty side instead.

Problem 5, page 5: rule out every smaller start

Three seeds begin with at most 12 exposed sides, while a full 4×4 board has 16. Four seeds begin with at most 16, while a full 5×5 board has 20. Growth cannot increase the count, so both are impossible. Adjacent seeds have even less starting perimeter; $4s$ is an upper bound, not an equality for every arrangement. The same comparison proves $s \geq n$ generally. A failed search alone would not establish that lower bound.

Indispensability, sources and later questions

Grades 4–5 Problem 6, page 6

The displayed four-seed set and deletion outcomes are the same as Grades 2–3 Problem 5. All four are needed for that start, but three can fill from another start. One different indispensable four-seed set is its quarter-turn:

$$\{(1,3),(2,3),(3,1),(3,2)\}.$$

Rotating a legal growth sequence preserves side-neighbors and the open square boundary, so this really does preserve indispensability. Children may invent another example; the checker finds 9 inclusion-minimal four-seed starts on 3×3, counting rotated/reflected placements separately. There are 14 three-seed filling starts. These counts are adult verification information, not extra printed enumeration demands.

Two notions worth keeping separate

In an inclusion-minimal start, deleting one seed destroys its success. That does not compare it with starts in different positions. A minimum start compares *all* positions and has the fewest possible seeds. Every minimum filling start is inclusion-minimal, since a removable seed would make a smaller filling start. The converse fails by the printed four-seed example. This distinction also appears in networks, generating sets and optimization beyond these boards.

A later question that changes the investigation

After the core work, ask how the minimum changes on an open $a \times b$ rectangle. Perimeter gives $s \geq \lceil (a+b)/2 \rceil$, but a lower bound alone does not supply a construction. Treat any rectangle result as a new conjecture or verified instance until it is proved; this packet claims the exact general theorem only for squares. Changing to a torus, counting diagonal neighbors or making occupied squares empty creates a different process and invalidates the stated perimeter argument or hypotheses. Do not change those rules during a test of a conjecture.

Connections and source boundaries

Atlas AD-07 already uses monotone closure and order-independent completion. Week 26 already counts exposed sides. Here they support economical generation and the gap between a locally indispensable and globally smallest start. The model is not chip conservation, rotor routing or prescribed layer growth.

Robert Morris, *Minimal percolating sets in bootstrap percolation*, introduction pp. 1–2 and Theorem 1 p. 3, supplies the research context and distinction between minimum and inclusion-minimal sets. Our elementary proofs, chosen examples and activity sequence are independently presented. Full references and overlap boundaries are in `SOURCES.md`.

Use and evidence

Independent finite checks verify the printed starts and deletion tests. The arguments in this guide establish the universal statements. The source package includes reproducible checks and a build script; its QA record distinguishes page review and extracted-archive reconstruction from physical rehearsal. These materials are unpiloted. Record actual preparation time, whether partners enforce side-neighbors reliably, where children need adult decisions, and which questions they choose to continue.